

Malware Concepts: RAT (Remote Access Trojan)

The Trojan guide's Section 3 flagged this directly as one of the most common payload categories a Trojan delivers, and the Rootkit guide's Section 5 flagged it as one of the things a rootkit most often exists to protect — a RAT is where those two threads meet. It's the payload that turns a single successful deception (the Trojan) into **ongoing, interactive, hands-on-keyboard control** of the victim's system, for as long as it remains undetected (often thanks to the Rootkit guide's concealment techniques working alongside it).

1. What a RAT Actually Is — The Foundation

A Remote Access Trojan (RAT) is malware that provides an attacker with **full, interactive, remote control** over an infected system — functionally similar to legitimate remote administration software (TeamViewer, AnyDesk, or even the Evil-WinRM tool from your Active Directory series), except installed covertly and operated without the victim's knowledge or consent.

The single defining trait: PERSISTENT, INTERACTIVE REMOTE CONTROL.

Unlike a Worm (which just spreads) or a basic Trojan payload that might do one specific thing, a RAT gives the attacker an ongoing, general-purpose command channel — closer in practical capability to an actual remote shell/session than to a single-purpose piece of malware.

Delivery Relationship to the Earlier Guides

- RATs are **FREQUENTLY delivered as a Trojan's payload** (see the Trojan guide, Section 3's "Backdoor Trojan" category).
 - RATs are **FREQUENTLY protected/hidden by an accompanying rootkit** once installed (see the Rootkit guide, Section 5).
 - A RAT is **NOT itself a self-propagating category** — it typically relies on the same delivery mechanisms as any Trojan payload, distinct from the Worm guide's autonomous spread.
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2. Typical RAT Architecture

Main Components

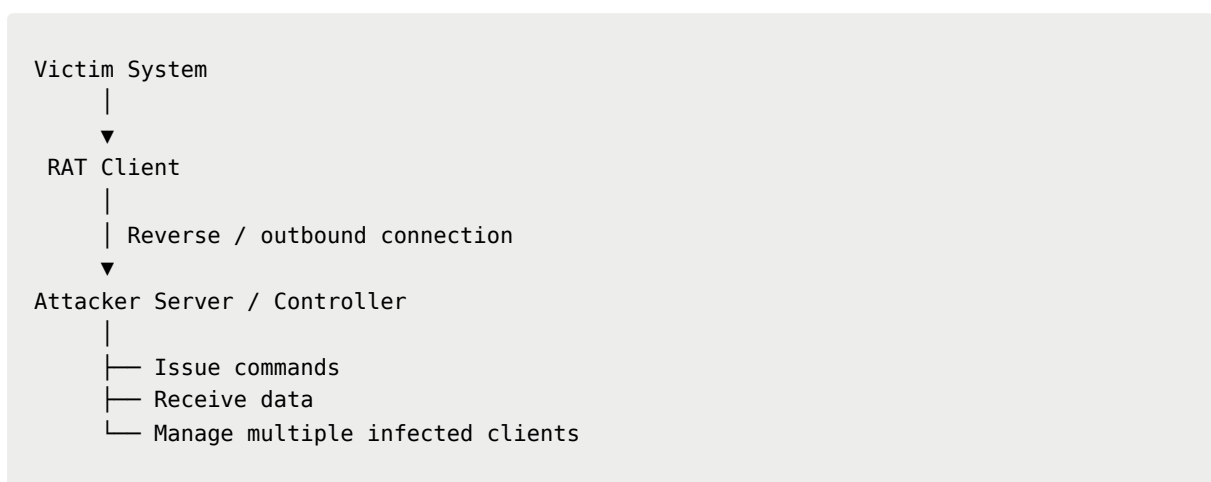
Component	Role
Client (on the victim's system)	The actual malware component, running silently, listening for or connecting out to instructions from the attacker.
Server/Controller (attacker-side)	The interface the attacker uses to issue commands and receive data from every currently-connected infected system — often supporting management of MANY infected clients simultaneously from a single control panel.

Connection Direction

Connection direction matters for evasion:

Connection Type	Description
Direct inbound connection	Attacker connects directly to the victim's listening port. This is often blocked by NAT/firewalls in real-world networks, making it less common in modern RATs.
Reverse/outbound connection	The INFECTED SYSTEM initiates the connection OUT to the attacker's server. This is the dominant modern approach, since outbound connections are far more likely to pass through firewalls and NAT unnoticed than unsolicited inbound ones.

Architecture Flow



3. Common RAT Capabilities

Capability	Function
File system access	Browsing, uploading, downloading, and deleting files on the infected system.
Command execution	Running arbitrary commands or scripts with the privilege level of the infected user/process.
Keystroke logging	Capturing everything typed, directly overlapping with the Spyware guide's core function.
Screen capture/live viewing	Taking screenshots or streaming a live view of the victim's screen.
Webcam/microphone access	Activating and capturing from connected hardware, without any visible indicator to the victim.
Credential harvesting	Extracting saved passwords, browser session data, or capturing credentials as they're typed.
Network pivoting	Using the infected system as a jumping-off point to reach further internal network resources — directly relevant to lateral movement concepts from your Active Directory series.

4. RAT Persistence Mechanisms

Because a RAT's entire value depends on remaining connected and available across system reboots, persistence is a core design concern, and frequently overlaps directly with techniques from your Linux Privilege Escalation series applied in reverse (as an installation goal rather than an exploitation target):

- **Registry Run keys / startup folder entries (Windows)**
- **Scheduled tasks / cron jobs** (directly connecting to the Cron Jobs guide's persistence angle) configured to relaunch the RAT client automatically
- **Service installation**, so the RAT runs with SYSTEM-level privilege and starts automatically at boot
- **Rootkit-based concealment** of any of the above persistence mechanisms, specifically to prevent a system administrator from simply noticing and removing an unfamiliar startup entry

5. Real-World Impact

Impact	Description
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Ongoing, general-purpose compromise rather than a one-time event	Unlike a single-purpose payload, a successfully installed and undetected RAT gives an attacker the flexibility to pursue WHATEVER goal becomes valuable over time — espionage, financial theft, further network compromise — without needing a fresh exploitation attempt each time.
Espionage and surveillance	The webcam/microphone and keystroke logging capabilities make RATs a preferred tool for targeted surveillance operations, distinct from broad, opportunistic malware campaigns.
Lateral movement enabler in enterprise environments	A single RAT-infected workstation can serve as the starting point for a much larger domain compromise, directly feeding into the attack chains covered throughout your Active Directory series.

6. Detection Considerations

- **Outbound network connections to unfamiliar or suspicious destinations**, particularly persistent, regularly-beaconing connections, are one of the strongest RAT indicators given the reverse-connection architecture described in Section 2.
- **Unexpected activation of webcam/microphone hardware indicators** (where the OS/hardware provides one) warrants immediate investigation.
- **Unusual process behavior** — a process with no legitimate reason to make outbound network connections doing so regularly — is a strong behavioral indicator, particularly when combined with persistence-mechanism review (Section 4).
- **Endpoint detection and response (EDR)** tooling with behavioral analysis is generally far more effective than static signature matching, given how easily RAT client binaries can be recompiled/modified to evade signature detection.
- **Cross-reference any newly discovered persistence mechanism** (scheduled task, service, startup entry) against a known-good baseline, exactly as recommended in your Cron Jobs guide's detection methodology.

7. Prevention

- **Address the DELIVERY mechanism directly** — since RATs are overwhelmingly delivered via Trojan-style deception, the Trojan guide's prevention recommendations (user awareness, application allowlisting, verified software sources) apply with equal force here.
- **Monitor and restrict outbound network connections** at the firewall/proxy level, specifically watching for the beaconing pattern characteristic of reverse-connection RAT architecture.
- **Deploy EDR with behavioral detection capability** specifically tuned to common RAT persistence and command-execution patterns.
- **Apply least-privilege principles broadly**, limiting the practical damage a RAT can do even when installed, by constraining the privilege level of the account it initially compromises.
- **Physically or logically disable unused webcam/microphone hardware** where not operationally required, directly closing Section 3's surveillance capability at the hardware level.
- **Segment networks** to limit the lateral movement value of any single RAT-infected host, directly reducing the severity described in Section 5's enterprise-compromise scenario.

Quick Reference Cheat Sheet

Defining Trait

Persistent, interactive, general-purpose **REMOTE CONTROL** — not a single-purpose payload.

Architecture

Client (on victim) ↔ Server/Controller (attacker side)

Modern RATs favor REVERSE/outbound connections to evade firewall/NAT restrictions.

Common Capabilities

File access, command execution, keystroke logging, screen/webcam/microphone capture, credential harvesting, network

pivoting

Typical Delivery & Protection Chain

Across this series:

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Trojan (delivery)
  ↓
RAT (payload / ongoing access)
  ↓
Rootkit (concealment, often protecting the RAT specifically)
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Persistence Mechanisms

Registry Run keys, startup folders, scheduled tasks/cron jobs, malicious services — often rootkit-concealed
